

WILL

Relational Geometry

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Einstein on Emergence of Spacetime

"There is no such thing as an empty space, i.e., a space without field. . . . Space-time does not claim existence on its own, but only as a structural quality of the field."

— *Albert Einstein*

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↓ 3 R as Philosophical Pillars: Rationalism, Operational Realism, Relationalism ↓			
Method: ↓(Epistemic Hygiene+Relational Origin+Ontological Minimalism+Math Transparency)↓			
Standard Primitives:		$\underbrace{\text{background manifold} + \text{metric}}_{\text{spacetime structure (not derived)}} + \underbrace{\text{G, c, M, fields}}_{\text{energy dynamics (postulated)}}$	
↓ (Ontological reduction by Relational Origin and Minimalism Principles no background allowed) ↓			
One Primitive:		$WILL \equiv SPACE-TIME-ENERGY \implies \boxed{\text{SPACETIME} \equiv \text{ENERGY}}$	
↓ All physically meaningful quantities are relational features of WILL ↓			
Topology of WILL: Relational Closure + Causal Continuity + Isotropy Theorems			
↓ Geometry of WILL: S^1 (1-DOF kinematic Push) and S^2 (2-DOF potential Pull) ↓			
Relational Conservation Theorem:		$\underbrace{\text{Amplitude}^2}_{\text{External Transformation}} + \underbrace{\text{Phase}^2}_{\text{Internal Change}} = 1$	
$\underbrace{1/\beta_Y = 1 + z_\beta}_{\text{transverse Doppler shift}}$	$\underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{Kinematic Push } S^1}$	Relational Projections	$\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{Potential Pull } S^2}$ $\underbrace{1/\kappa_X = 1 + z_\kappa}_{\text{gravitational redshift}}$
↓ (by DOF-Indifference Lemma: equal quadratic weight to each independent DOF) ↓			
Closure Theorem:		$\underbrace{1 \text{ PULL}^2}_{\text{Amplitude on } S^2} = \underbrace{2 \text{ PUSH}^2}_{\text{Amplitude on } S^1} \implies \boxed{\kappa^2 = 2\beta^2}$	
↓ (by Relational Conservation Theorem derive metric intervals as scaffolding) ↓			
Minkowski:	$\underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{irreducible primitive}}$	$+ \underbrace{x, y, z, t...}_{\text{coordinate inflation}}$	$= \underbrace{c^2 d\tau^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2}_{\text{mathematical scaffolding}}$
Schwarzschild:	$\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{irreducible primitive}}$	$+ \underbrace{x, y, z, t...}_{\text{coordinate inflation}}$	$= \underbrace{c^2 d\tau^2 = c^2(1 - (R_s/r)) dt^2}_{\text{mathematical scaffolding}}$
↓ (by Relational Closure and Causal Continuity Theorems: any change must be balanced) ↓			
Energy Symmetry Law:		$E = E_0 \kappa_X / \beta_Y \longrightarrow \Delta E_{A \rightarrow B} + \Delta E_{B \rightarrow A} = 0$	
$\Delta E_{A \rightarrow B} = E_0(\kappa_{X,B} / \beta_{Y,B} - \kappa_{X,A} / \beta_{Y,A}); \Delta E_{B \rightarrow A} = E_0(\kappa_{X,A} / \beta_{Y,A} - \kappa_{X,B} / \beta_{Y,B})$ ↓			
Speed of light: $\beta \rightarrow 1 \rightarrow \beta_Y \rightarrow 0 \rightarrow E = E_0 \cdot \kappa_X / 0 \rightarrow \infty \rightarrow$ requires ∞ energy			
Event Horizon: $\kappa \rightarrow 1 \rightarrow \kappa_X \rightarrow 0 \rightarrow E = E_0 \cdot 0 / \beta_Y \rightarrow 0 \rightarrow$ requires 0 energy			
by Unified Relational Scaling Lemma: Equivalence Principle		$m_g \equiv m_i \equiv m = E_0 / c^2$	
↓ (Lagrangian and Hamiltonian derived as collapsed single-point approximations) ↓			
$\underbrace{\Delta E_{A \rightarrow B} + \Delta E_{B \rightarrow A} = 0}_{\text{exact symmetry law}}$		$\underbrace{\frac{1}{2}(\kappa_A^2 - \kappa_B^2) + \frac{1}{2}(\beta_B^2 - \beta_A^2)}_{\text{first order approximation}} \approx \underbrace{T \pm V}_{\text{ontological collapse}}$	
↓ Relational Orbital Mechanics		R.O.M.: a closed algebraic system of equations ↓	
↓ Schwarzschild Metric $\tau_o^2 = \kappa_{X_o}^2 - (\kappa_{X_o}^2 \beta_R)^2 \kappa_{X_o}^{-2} - (\kappa_{X_o} \beta_T)^2$ as Scaffolding ↓			
$\underbrace{\tau_o^2 = \kappa_{X_o}^2 \beta_{Y_o}^2}_{\text{irreducible primitive}}$	$+ \underbrace{x, y, z, t...}_{\text{coordinate inflation}}$	$= \underbrace{\left(\frac{d\tau}{dt}\right)^2 = (1 - R_s/r) - \frac{(dr/cdt)^2}{(1 - R_s/r)} - \left(\frac{rd\phi}{cdt}\right)^2}_{\text{mathematical scaffolding}}$	
↓(by Factorization Theorem: every quantity X splits into unitless core $\hat{X}$ and scale label S_X)↓			
Unitless Core:		$X = \underbrace{\hat{X}(\beta, e)}_{\text{unitless core}} \cdot \underbrace{S_X(R_s, c, G)}_{\text{units-scale}}$ G and M are unit-conversion factors.	
↓ (by Inverse Square, Closure and Causal Continuity Theorems) ↓			
No Singularities: $\beta_{\max}^2 = 1 \rightarrow \kappa_{\max}^2 = 2 \rightarrow r_{\min} = R_s / \kappa_{\max}^2 \rightarrow r_{\min} = R_s / 2$			
The point $r = 0$ is absent from the admissible relational range			
↓ (derive Relational Field Equation:) ↓			
$\boxed{\kappa^2 = R_s / r = \rho_{\text{field}} / \rho_{\max}}$		$\implies \boxed{\text{Spacetime Geometry}(R_s / r)} \equiv \boxed{\text{Energy Density}(\rho / \rho_{\max})}$	
↓ Theoretical Ouroboros ↓			
$\boxed{\text{SPACETIME} \equiv \text{ENERGY}}$		$\Leftrightarrow \boxed{\text{SPACETIME GEOMETRY} \equiv \text{ENERGY DENSITY}}$	

This document must be read **literally**.

- All terms are defined within **Relational Logic**: *Positions, velocities and other properties of objects are only meaningful relative to other objects. Space without objects is operationally undetectable and physically meaningless, the same way time is meaningless without events. Within relational logic space and time are systems of relations that exist between objects.*
- Any attempt to reinterpret these derivations through conventional substantival metric notions (*absolute distances, external backgrounds, hidden containers*) will produce distortions and mathematical misreadings.
- **Take the words as written** and ontology as derived, without importing external concepts.

Remark 0.1 (The term "geometry"). *Is used herein in its fundamental, classical sense: the pure logic of relations, ratios, and structural constructions. It **does not** refer to modern differential geometry. There is no a priori metric tensor ($g_{\mu\nu}$), no Riemannian manifold, and no underlying coordinate grid. The geometry emerges from physical relations, not from a background space.*

1 Stage I: Ontology Emerges from Epistemology

"There is no such thing as philosophy-free science; there is only science whose philosophical baggage is taken on board without examination."

Daniel Dennett (from Darwin's Dangerous Idea, 1995)

Every logical step is subject to empirical audit: if any consequence of this chain contradicts observation, the chain fails.

Remark 1.1 (On mirroring effect). *As humans, we often project ourselves onto observed phenomena, obscuring the true nature of physical reality behind our anthropic tendencies. In order to prevent this mirroring effect, theoretical development has to establish strong epistemic constraints on scientific reasoning. Without this constrains our resulting theories risking telling us more about ourselves than about the nature of the Universe.*

1.1 Foundational Epistemic Principles

Flow chart of derivation logic

[Logos Map](#)

The foundations of this research bridge philosophy and practice, mapping three core philosophical pillars directly to our operational methodology:

- **Rationalism**

Principle 1.2 (Epistemic Hygiene). Refusal to import unjustified assumptions or arbitrary choices.

This line of reasoning derives physics by **removing hidden assumptions**, rather than introducing new postulates. This construction is deliberate and contains zero free parameters. This is not a simplification — it is a deliberate epistemic constraint. No assumptions are introduced and no constructs are retained unless they are geometrically or energetically necessary.

Principle 1.3 (Ontological Minimalism). Maintain the minimum number of foundational primitives by peeling off interpretational layers until only the irreducible core is left. *Spatial container, axis of time, distance, forces, fields, and intrinsic isolated properties are excluded as preexisting primitives.*

These traditional terms (time, distance, forces, fields) later emerge naturally as translations into a common descriptive vocabulary.

- **Operational Realism**

Principle 1.4 (Mathematical Transparency). *Every mathematical expression, phrase, operation, or identity is treated as a binding ontological statement. This forces mathematics to always reflect direct physical meaning, preventing obscure constructs. Every symbol must be anchored to a unique physical idea with a transparent ontological origin.*

This principle transforms mathematics from merely an arbitrary toolbox used at whim into a rigorous language where every single element must be ontologically justified by internal derivation or direct, external empirical measurement. Introducing symbols without explicit necessity constitutes **over-parameterization**: the proliferation of symbols without corresponding physical meaning, governed by the equivalence:

- **Relationalism**

Principle 1.5 (Relational Origin). As a default epistemic stance, all physical quantities are defined exclusively by their relations, until and unless their absolute nature is empirically proven.

Any introduction of unproven absolute properties risks reintroducing metaphysical artifacts and directly contradicts the foundational insight of relationalism.

A detailed exploration of these philosophical roots is beyond the scope of this paper, which remains focused on an empirically measurable physical reality.

A Note on Methodology vs. Ontology

It may be objected that these foundational principles constitute a set of axioms, thereby contradicting the claim of *zero assumptions*. This objection, however, conflates *methodological constraints* with *ontological axioms*.

Under the strictures of the scientific method, the burden of proof rests exclusively on the claimant asserting an absolute existence. Refusing to import unobserved absolutes (such as a background container or intrinsic properties) is not an assumption; it is the default epistemic stance — the null hypothesis.

These principles function as rules of logical engagement designed to prevent the introduction of surplus ontology. To challenge them is to argue for the active admission of a specific, unobserved absolute; under the empirical foundations of physics, the burden of proof rests entirely with that admission.

1.2 Ontological Reduction

Without background "container" and without introducing new primitives we should see structure and dynamics as a single ontological element 1.6. This leads to Ontological Reduction.

Lemma 1.6 (False Separation). *Any model that treats processes as unfolding within an independent background necessarily assigns to that background structural features (metric, orientation, or frame) not derivable from the relations among the processes themselves. Such a background constitutes an extraneous absolute.*

Proof. Suppose an independent background exists. Then at least one of its structural attributes - metric relations, a preferred orientation, or a class of inertial frames - remains fixed regardless of inter-process data. This attribute is not relationally inferred but posited a priori. It thereby violates the relational closure principle: it introduces a non-relational absolute external to the system. Hence the separation is unwarranted. $\square$

Corollary 1.7 (Irreducible Primitive). *Without introducing new primitives 1.6, we should see structure and dynamics as a single ontological element. This leads to Ontological Reduction to a Single The structural arena and the dynamical content must be identified: structure is dynamics, energy is spacetime.*

$$\boxed{SPACETIME \equiv ENERGY}$$

This leads us to a single irreducible primitive with *negative ontological weight*: it reduces the number of primitives and removes the separation between structure and dynamics. Spacetime is not a "container" but relational tension between energy states.

Theorem 1.8 (**WILL**). *The Single Irreducible Fundamental Primitive is the unified relation structure of space-time-energy that we call **WILL** $\equiv$ **SPACE-TIME-ENERGY**.*

By Ontological Reduction. By Ontological Minimalism Principle (1.3) and False Separation Lemma (1.6) spacetime and energy can not be seen as separate irreducible ontological primitives. This leads to ontological reduction in to a unified relation structure of space-time-energy. **WILL** $\equiv$ **SPACE-TIME-ENERGY** is the technical term we use for this structure. $\square$

Remark 1.9 (Auditability). *The WILL Theorem 1.8 is foundational but testable. it is subject to: (i) geometric audit (internal logical consequences) and (ii) empirical audit (agreement with empirical data).*

Definition 1.10 (Ontological weight). *The number of irreducible primitives a theory posits.*

<i>Picture</i>	<i>Primitives</i>
<i>Substantial: container + content (spacetime, energy/matter)</i>	≥ 2
<i>Relational: WILL (one self-relating structure)</i>	1

This Ontological Reduction is the operation physics has performed at every unification ??. Each merged two primitives into one. **SPACETIME** $\equiv$ **ENERGY** applies the identical operation. **WILL does not add, it subtracts: it removes the hidden assumptions. Structure and dynamics are two aspects of a single primitive that we call WILL.**

Features of WILL

All physically meaningful quantities are relational features of WILL.

1.3 The Foundational Core: Topology of WILL

Definition 1.11 (Closed System). *The term **Closed System** we use to describe a complete accounting of all relevant relational channels. No input or output left unbalanced. The system is fully self-contained.*

Theorem 1.12 (Relational Closure). *A purely relational system cannot be geometrically open; any stable, self-contained system must be described by a closed and self-consistent relational geometry. Its geometry is closed by absolute analytic necessity.*

Proof by Analytic Tautology.

1. **Premise 1 (Relational Origin):** Within RG Physical quantities are defined exclusively by their relations (Principle 1.5).
2. **Premise 2 (Interaction as Integration):** If a hypothetical “outside” entity or boundary were to physically interact with the system, that interaction itself constitutes a relation.
3. **Deduction (The Tautology):** As relation is formed, it is mathematically and physically integrated into the system.

An “open relational system” experiencing external influence is a logical contradiction. The system is closed not by a spatial boundary or a metric container, but by the strict impossibility of an interacting non-relation. The system is relationally closed by definition. $\square$

1.4 What is Energy in Relational Ontology?

Across all domains of physics, one empirical fact persists: in every closed system there exists a quantity that never disappears or arises spontaneously, but only transforms in form. This invariant is observed under many guises — kinetic, potential, thermal, chemical — yet all are interchangeable, pointing to a single underlying structure.

Crucially, this quantity is never observed directly, but only through *differences between states*: a change of velocity, a shift in configuration, a transition of phase. Its value is relational, not absolute: it depends on the chosen frame or comparison, never on an object in isolation.

Moreover, this quantity provides continuity of causality. If it changes in one part of the system, a complementary change must occur elsewhere, ensuring the unbroken chain of transitions. Thus it is the bookkeeping of causality itself — the ledger of change.

Corollary 1.13 (Energy). *In classical and relativistic mechanics, this derived invariant measure of change is assigned the historical label “Energy”. By showing its relational origin, we strip “Energy” of its substantialist properties (as a “substance” or “thing”) and reveal its origin as the necessary measure of change and bookkeeping ledger of causality.*

Energy :

Definition 1.14 (Energy).

*Energy - relational measure
of state transformation
or difference between states.*

Remark 1.15 (Energy Ledger: intuitive understanding). *Consider a completely closed economy with no physical currency — no dollar bills, no gold coins, and no central bank (acting as an absolute container). It is a purely relational economy consisting solely of a massive ledger of debts and credits between individuals. If one account decreases, another must increase. If the system is frozen at any given moment, the sum of all debts and credits across the entire ledger must equal zero. “Money” does not exist as a physical object; it is simply the mathematical rule that the ledger must always balance. Energy functions identically: it is not a substance, but the invariant rule of relational accounting.*

Theorem 1.16 (Conservation and Causal Continuity). *Without background any closed dynamical system must conserve its total energy. Any energy change must be perfectly balanced by a complementary change elsewhere in the system enforcing **Conservation and Causal Continuity**.*

Proof by Structural Necessity.

1. **Premise 1 (Relational Origin):** Physical quantities are defined exclusively by their relations. No background allowed (Principle 1.5).
2. **Premise 2 (Empirical Change):** States undergo observable Change.
3. **Premise 3 (Relational Closure):** As proven in Theorem 1.12, the relational system is analytically closed. There is no external background, void, or hidden container to absorb or inject influence.

4. **Deduction (Causal Continuity):** Because the system is closed (Premise 3), and no background allowed (Premise 1), an empirical change (Premise 2) cannot vanish into a void or emerge from one. To maintain the structural integrity of the relations (Premise 1), any change must be perfectly balanced by a complementary change elsewhere in the system. This enforcing **Conservation and Causal Continuity**.

□

Theorem 1.17 (Relational Isotropy). *Within a background-free relational system, the energy state has no direction or coordinate that can be a priori privileged. Therefore, the geometry encoding the relational change (energy) must be maximally symmetric (isotropic).*

Proof by Relational Origin.

1. **Premise 1 (Absence of a Grid):** As established by Theorem 1.12 (Relational Closure), the relational system possesses no external container, absolute metric tensor, or underlying coordinate grid.
2. **Premise 2 (Direction requires Relation):** Under the Principle of Relational Origin (Principle 1.5), “direction” or “orientation” (not only spatial) cannot exist in a void; it only physically manifests as a specific interaction or measurement between defined participants.
3. **Deduction (Pure Gauge):** Because there is no absolute grid (Premise 1), assigning an intrinsic, absolute axis to the unmeasured relational ledger (energy) introduces unobservable “surplus ontology.” Any such arbitrary orientation is a mathematical artifact (pure gauge).

Therefore, prior to a specific interaction defining a local axis, the pure relational structure encoding this conserved energy ledger must possess no intrinsic privileged direction. It must be isotropic. □

2 Stage II: Relational Geometry of WILL

2.1 WILL Structure

Having established the Single Unifying Primitive 1.8 by removing the unwarranted separation of structure and dynamics, and its topology, now we proceed to derive the Relational Geometry of **WILL**. The foundational core (1.3) — Closure, Causal Continuity, and Isotropy — establishes the necessary constraints.

Relational Carrier Dictionary:

All references to “carriers” within WILL RG open research are to be read in the strict relational sense:

- **Degree of freedom (DOF):** The minimum number of independent relational variations required to encode a physically distinguishable, observable Change. DOF counts the irreducible relational structure required by the phenomenon being represented; it is not an independently posited geometric dimension.
- **Direction:** An oriented relation in state space. Opposite orientations are physically distinct and not identified.
- **Closed carrier:** Compact and without boundary; relational ledger cannot leak into an external void.
- **No background:** No external embedding state space. All geometric structure is reconstructible solely from relations between participants.
- **Maximal symmetry:** The carrier is homogeneous and isotropic with respect to oriented directions.
- **Minimal relational carrier:** A closed, maximally symmetric carrier containing exactly the minimum DOF required to represent the corresponding observable relational Change, without arbitrary additional structure.

2.2 Deriving Relational Carriers

Theorem 2.1 (Minimal Relational Carriers of the Conserved Relational Ledger). *The minimal relational carriers of the same conserved energy ledger satisfying methodological constraints (1.1) and foundational core (1.3) of Relational Closure, Conservation, Maximal Symmetry Theorems are:*

- (a) S^1 for the minimal 1-DOF directional (Kinematic) relational Change;
- (b) S^2 for the minimal 2-DOF omnidirectional (Potential) relational Change.

Proof. The carriers are determined by the minimum irreducible DOF required to represent observable relational Change, with no assumption of additional DOF beyond what observation requires.

1. 1-DOF (Kinematic) relational Push.

The simplest non-trivial physically distinguishable Change is a difference between two relational states, A and B . Its smallest non-zero irreducible relational structure has one degree of freedom.

This 1-DOF Change can be represented as a closed line or loop in relational state space. The minimal closed, maximally symmetric carrier is therefore the circle S^1 . The total Change ledger on S^1 we normalise to unity making S^1 - unit circle. Relational state correspond to the point on S^1 we call θ_1 with two orthogonal external and internal projections represent two aspects of the same 1-DOF relational Change:

$$\boxed{\beta^2 + \beta_Y^2 = 1}.$$

The projections β (external amplitude) and β_Y (internal phase) do not constitute two independent DOF; they are orthogonal projections of the same single relational degree of freedom.

The carrier is therefore:

- **Closed** (by Relational Closure Theorem 1.12 — no boundary)
- **Maximally symmetric** (by Isotropy Theorem 1.17 — no privileged point or direction on the carrier)

A linear segment cannot carry this relation because it possesses absolute endpoints. Endpoints act as boundaries where the transformational ledger would either vanish (violating the Conservation Theorem) or require an external bounding mechanism to reflect it (violating the Relational Origin and Ontological Minimalism principles 1.3, 1.5).

The unique 1-DOF Push carrier is the circle (S^1)

2. 2-DOF (Potential) relational Pull.

The next simplest irreducible non-trivial physically distinguishable Change is 2 DOF relational structure. It arises when a central state A is related by the same Change to N equivalent states B_i , with no privileged direction among those relations. It is an isotropic omnidirectional relation.

The 2-DOF carrier must be:

- **Closed** (by Relational Closure Theorem 1.12 — compact, no boundary, no leakage into an external void.)
- **Maximally symmetric** (by Isotropy Theorem 1.17 — isotropic distribution of the relational Change.)
- **Orientable** (by Causal Continuity Theorem 1.16 — opposite directions are physically distinct, not identified)

Under these conditions, the minimal boundaryless, maximally symmetric 2-DOF carrier is the 2-sphere (S^2). Anisotropic surfaces such as the torus T^2 introduce additional directional structure, while the non-orientable $\mathbb{RP}^2$ identifies opposite orientations and therefore does not satisfy foundational core (1.3) .

Meridional Section of S^2

By Isotropy Theorem 1.17, any radially symmetric relation reduces to a great-circle meridional section. The S^2 carrier remains the physical/relational carrier; the calculation is restricted to a symmetry-equivalent meridional section without loss of generality.

Same as S^1 the total relational Change ledger on S^2 we normalise to unity making S^2 a unit two-sphere. Relational state correspond to the point on meridional section of S^2 we call θ_2 with two orthogonal external and internal projections represent two aspects of the same 2-DOF relational Change:

$$\boxed{\kappa_X^2 + \kappa^2 = 1}.$$

The projections κ_X (internal phase) and κ (external amplitude) are not separate physical quantities; they are orthogonal projections of the same 2-DOF relational Change.

The unique 2-DOF Pull carrier is the 2-sphere (S^2)

3. Higher-Dimensional Carriers (S^n for $n \geq 3$).

The existence of mathematical carriers with $n \geq 3$ DOF does not by itself justify their ontological inclusion. DOF is introduced only when an observable phenomenon requires an irreducible relational structure beyond those already sufficient.

No observable phenomenon addressed in the present research requires a 3-DOF carrier. Therefore no 3-DOF or higher carrier is introduced at this stage.

This is not a claim that higher relational DOF do not exist. If future empirical observations require an additional irreducible relational degree of freedom, the carrier structure must be extended accordingly. Until such empirical necessity appears, introducing additional DOF would add structure without an identified physical relation and would violate Principle 1.3 and Principle 1.4.

Ontological Minimalism and Mathematical Transparency therefore retain S^1 and S^2 as the minimal relational carriers required by the presently expressed observable phenomena, without invoking metric geometry *a priori*. The possibility of additional carriers remains an open empirical question. □

- 1-DOF: minimum irreducible relational structure required to encode a directional Push (kinematic)
- 2-DOF: minimum irreducible relational structure required to encode an isotropic omnidirectional Pull (potential)
- 3-DOF+: not introduced unless empirical observations require an additional irreducible relational degree of freedom

WARNING: The Spatial Container Fallacy

S^1 and S^2 are not to be interpreted as physical geometries embedded in a spacetime container.

Do not visualise a sphere or a ring expanding through a 3D void.

- S^2 has no physical surface area, no volume, and does not "dilute" a flux across a spatial distance.
- S^1 and S^2 are algebraic state-space carriers encoding relational structures of the same conserved energy ledger.

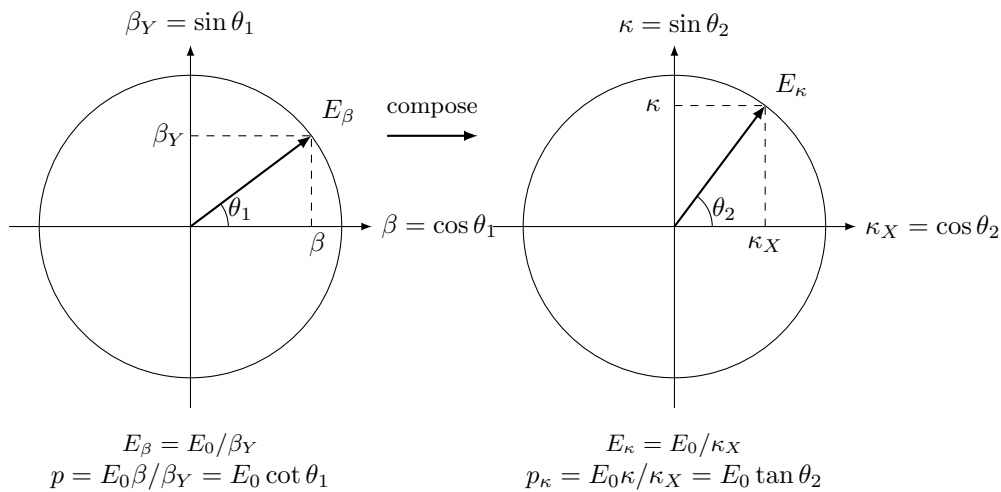
Relational Carriers S^1 and $S^2 \equiv$ Logic of the Proportions

Spatial distance (r) and time (t) are emergent descriptive labels that observers attach to these underlying algebraic phase ratios. The phase projection dictates the space; the space does not dictate the phase.

Remark 2.2 (Constructive Derivation Protocol). *This derivation proceeds by construction from traced origins. Each object entering the chain — the principles, the theorems, and the carriers — has an explicit derivation path. Alternative structures (discrete groups, higher-dimensional carriers, non-orientable surfaces) are not excluded by assumption; they are excluded only when they do not satisfy the stated relational constraints or are not required by the presently observable phenomena. Higher-dimensional carriers remain admissible should future empirical evidence require additional irreducible relational DOF.*

2.3 The Amplitude-Phase Duality

S^1 and S^2 encode the *ratio* of internal to external relational energy (state transformation ledger 1.4). A point on S^1 and S^2 is not a position but a partition: how much of the system's transformation is externally expressed (amplitude β, κ) versus internally retained (phase β_Y, κ_X). The carrier geometry makes this partition exact and parameter-free.



Lemma 2.3 (Duality of Relation). *By the Principle of Relational Origin (1.5), the physical manifestation of a state cannot be quantified by a single absolute magnitude. The elimination of background manifolds necessitates a complementary internal ratio of two measures:*

1. the **amplitude** of transformation (external shift), and
2. the **phase** of transformation (internal order).

Proof. By the Principle of Relational Origin (1.5), assigning a single unbounded scalar to describe a transformation introduces an unobservable absolute reference scale. A purely relational state must therefore be defined internally, as a ratio or partitioning of a closed resource.

The geometric carriers S^1 and S^2 mathematically enforce this requirement. Identifying a unique state on these closed carriers necessitates two orthogonal projections. A single projection cannot capture a ratio. One projection must quantify the extent of the external relation (Amplitude), which geometrically dictates an orthogonal complement quantifying the retained internal structure (Phase). Together, these dual, coupled axes provide the minimal, parameter-free ratio required to define a state transformation without invoking an external background. $\square$

The orthogonal decomposition of the relational carriers S^1 and S^2 reveals a functional duality. Physical state is a combination of two projections:

Definition 2.4 (The Amplitude Projection (External Interaction)). Denoted by β (kinematic) and κ (potential). This component measures the extent of external relation. It manifests as momentum or potential intensity. Physically, it represents the system's relational "shift" from the **Observer's Relational Origin** (the rest frame).

$$\text{Amplitude} \rightarrow \text{External Power (Kinetic/Potential)}$$

Definition 2.5 (The Phase Projection (Internal Evolution)). Denoted by β_Y and κ_X . This component measures the internal ordering. It governs the intrinsic scale of proper time and proper length. A Phase of 1 represents maximal internal flow (rest), while a Phase of 0 represents the collapse of internal causality (light-speed/event horizon).

$$\text{Phase} \rightarrow \text{Internal Order (Time/Structure)}$$

Theorem 2.6 (Conservation of Relation). For both kinematic (S^1) and potential (S^2) modes, the sum of the squared Amplitude and squared Phase is invariant:

$$\underbrace{\text{Amplitude}^2}_{\text{External Interaction}} + \underbrace{\text{Phase}^2}_{\text{Internal Existence}} = 1 \quad (1)$$

Proof. By Theorem 2.1, the balanced relational ledger is carried by the maximally symmetric geometries S^1 and S^2 . Normalizing the total relational ledger to unity (1), any physical state corresponds to a point on these carriers. The algebraic identity of orthogonal projections on a unit circle and two-sphere dictates that the sum of their squares must equal the squared radius. Therefore, the parameters satisfy $\beta^2 + \beta_Y^2 = 1$ and $\kappa^2 + \kappa_X^2 = 1$, encoding the finite, closed relational ledger. $\square$

2.3.1 Consequence: Relativistic Effects

Proposition 2.7 (Physical Interpretation: Relativistic Effects). The Conservation of Relation Theorem 2.6 implies that any redistribution between the orthogonal components manifests physically as the relativistic effects of time dilation and length contraction.

Proof. By Theorem 2.6, the components satisfy $\beta^2 + \beta_Y^2 = 1$ and $\kappa^2 + \kappa_X^2 = 1$. An increase in Amplitude (β, κ) enforces a geometric decrease in Phase measure (β_Y, κ_X). This necessary reduction of β_Y and κ_X corresponds to the dilation of proper time and the contraction of proper length, while the growth of β and κ represents momentum or potential depth. Thus, the kinematic and potential trade-off is the direct, un-parameterized physical expression of the geometric closure of the S^1 and S^2 relational carriers. $\square$

This identifies Relativistic and Gravitational Time Dilation not as a mysterious "slowing down" of clocks, but as geometric **phase rotation**. As a system invests more of its relational energy ledger into external Amplitude (β or κ), it necessarily withdraws from its internal Phase (β_Y or κ_X), changing the rate of its proper time evolution.

2.4 Phase Projections as Spectroscopic Shifts

Theorem 2.8 (Spectroscopic Phase Shift). The potential Pull projection κ^2 is determined by the measurable gravitational redshift z_κ via the identity:

$$\kappa^2 = 1 - \frac{1}{(1 + z_\kappa)^2}. \quad (2)$$

Proof. By the Conservation of Relation Theorem, the gravitational Pull κ and internal phase κ_X satisfy:

$$\kappa^2 + \kappa_X^2 = 1 \implies \kappa^2 = 1 - \kappa_X^2. \quad (3)$$

By Duality of Relation Lemma (2.3) the phase projection governs the intrinsic flow of proper time, the frequency of emitted light scales by the internal phase. The observable spectroscopic shift z_κ is therefore the operational ratio between the observer's phase and the source's phase:

$$1 + z_\kappa = \frac{\kappa_{X(\text{obs})}}{\kappa_{X(\text{source})}}. \quad (4)$$

Substituting the observer's maximal phase ($\kappa_{X(obs)} = 1$) — representing the invariance of the local measurement against the local frame — the phase of the source is isolated:

$$\kappa_{X(source)} = \frac{1}{1 + z_\kappa}. \quad (5)$$

Substituting this internal phase back into the initial closure invariant yields the algebraic identity:

$$\boxed{\kappa^2 = 1 - \frac{1}{(1 + z_\kappa)^2}} \quad (z_\kappa = \text{gravitational redshift})$$

□

Theorem 2.9 (Kinematic Phase Shift (Transverse Doppler)). *The kinematic Push projection β^2 is determined by the transverse Doppler shift z_β via the symmetric identity:*

$$\beta^2 = 1 - \frac{1}{(1 + z_\beta)^2}. \quad (6)$$

Proof. By the S^1 closure invariant (Conservation Theorem), the kinematic amplitude β and internal kinematic phase β_Y satisfy:

$$\beta^2 + \beta_Y^2 = 1 \implies \beta^2 = 1 - \beta_Y^2. \quad (7)$$

The transverse Doppler shift z_β (without observer dependant line-of-sight kinematics) is the optical measure of phase discrepancy ratio:

$$1 + z_\beta = \frac{\beta_{Y(obs)}}{\beta_{Y(source)}}. \quad (8)$$

The observer's external kinematic projection is zero against local frame ($\beta_{obs} = 0$), yielding internal kinematic phase: $\beta_{Y(obs)} \equiv 1$. Substituting $\beta_{Y(obs)} = 1$, we isolate the internal phase of the moving source:

$$\beta_{Y(source)} = \frac{1}{1 + z_\beta}. \quad (9)$$

Substituting this back into the S^1 closure invariant yields the algebraic identity:

$$\boxed{\beta^2 = 1 - \frac{1}{(1 + z_\beta)^2}} \quad (z_\beta = \text{transverse Doppler shift})$$

□

Theorem 2.10 (Operational Measurability). *The relational projections are encoded in the combined phase interactions of light (spectroscopy) and are operationally independent of spatial distance r , mass M , and the gravitational constant G .*

Proof. Step 1: Total accumulated phase difference (τ). Spectroscopy does not measure "gravitational potential" or "kinematic velocity" as independent isolated vectors; it measures the total accumulated phase discrepancy between the source and the observer. We define the **Relational Spacetime Factor** τ as the inverse of the system's combined measured redshift:

$$\tau \equiv \frac{1}{Z_{sys}} = \frac{1}{(1 + z_\kappa)(1 + z_\beta)}. \quad (10)$$

This single observable represents the product of the internal phase projections of the carriers S^2 and S^1 :

$$\tau = \underbrace{\kappa_X}_{\text{Potential Phase}} \cdot \underbrace{\beta_Y}_{\text{Kinematic Phase}} = \sqrt{1 - \kappa^2} \sqrt{1 - \beta^2}. \quad (11)$$

Step 2: Relational Identity. Expanding the product of the squared phases yields the relationship between the optical signal τ and the geometric state:

$$\tau^2 = \kappa_X^2 \beta_Y^2 = (1 - \kappa^2)(1 - \beta^2) = 1 - (\kappa^2 + \beta^2) + \kappa^2 \beta^2. \quad (12)$$

The optical signal contains the complete information regarding the system's structural relational state, acquired directly through observation without invoking absolute properties. □

- **Push** S^1 is the minimal geometric structure that can encode a directional relation without absolute space. Its parameter is an angle θ_1 whose cosine ($\beta = \cos(\theta_1)$) we identify with the fraction v/c and whose sine ($\beta_Y = \sin(\theta_1)$) governs the internal ordering (proper spacetime fraction).
- **Pull** S^2 is the minimal geometric structure that can encode omnidirectional relation without absolute space. Its parameter is an angle θ_2 whose sine ($\kappa = \sin(\theta_2)$) we identify with the escape velocity fraction v_e/c and whose cosine ($\kappa_X = \cos(\theta_2)$) governs the internal ordering (proper spacetime fraction).

The unit radius of kinetic and potential carriers represents the Constant Universal Rate of Change, the Global Tempo of Transformations. The angles θ_1 and θ_2 control the distribution of this constant rate between the phase and amplitude components. *Any two observers regardless of their cultural differences (including hypothetical extraterrestrial intelligence) assign β and κ ratios the same value for the same state, whatever their unit system — or even their descriptive vocabulary. These mappings are NOT definitions:*

$$\beta = \frac{v}{c}, \quad \kappa = \frac{v_e}{c} = \sqrt{\frac{R_s}{r}} = \sqrt{\frac{\rho}{\rho_{max}}}$$

They are *translations* into descriptive local historical vocabulary. Relational projections are **Metrologically Independent**.

2.5 Relational Origin of Spatial Distance

Spatial distance is derived as the inverse of the potential projection. The familiar inverse-square force law appears only in the metric scaffolding of legacy coordinate descriptions and is not part of the relational core.

Theorem 2.11 (Inverse-Distance Potential Amplitude). *The inverse-distance potential proportionality $\kappa^2 \propto 1/r$ is a geometric necessity of the S^2 carrier's relational ledger, determined without spatial primitives.*

Proof. Distance cannot be an *a priori* foundational primitive. It violates the principles of Relational Origin (1.5) and Ontological Minimalism (1.3). Therefore it must be defined by the relational tension between energy states. By the [Spectroscopic Phase Shift Theorem](#), the potential projection κ^2 is operationally extracted entirely from the dimensionless redshift z_κ , independent of spatial parameters.

S^2 is the omnidirectional relational carrier with two degrees of freedom. Let κ be the active potential projection per independent degree of freedom. Since the omnidirectional ledger distributes identically over both DOF (2.13), the dimensionless radial measure $\tilde{r}$ is the inverse of the external amplitude projection counted once for each DOF:

$$\tilde{r} = \frac{1}{\kappa} \cdot \frac{1}{\kappa} = \frac{1}{\kappa^2}. \quad (13)$$

Saturation point $\kappa^2 = 1$ defines causal horizon $r = R_s$ 2.12 and spatial distance emerges as

$$r = \frac{R_s}{\kappa^2}, \quad \Longleftrightarrow \quad \kappa^2 = \frac{R_s}{r}. \quad (14)$$

Thus $\kappa^2 \propto 1/r$. No background manifold, coordinate derivative, or force law is introduced. This is a direct realization of the Unifying Primitive (1.8). The $1/r$ proportionality is an analytic identity defining spatial distance, not an empirical input. $\square$

Remark 2.12. *Here and in all RG papers the causal horizon $\kappa = 1$ is symbolised with the usual letters for the Schwarzschild radius R_s for translational ease. Despite their ontological difference, these two horizons are mathematically identical. In order to keep the number of newly introduced symbols at a minimum, the historical symbols remain. The complete independence of causal horizon $\kappa = 1$ from its historical origin $R_s = \frac{2GM}{c^2}$ is proven in [R.O.M. sections sec:unitless core, sec:relational_parameterization, sec:absolute_scale, sec:Kerr](#) and others.*

Summary

The inverse-square proportionality does not exist because physical space dictates how forces dilute. It exists because “spatial distance” is the descriptive vocabulary assigned to the inverse amplitude of the potential relational carrier S^2 .

The geometry of spacetime is dictated by the Relations of energy states.

2.6 Closure: Relational Origin of the Virial Theorem

Having established that directional (kinematic) and omnidirectional (potential) relations are carried by the S^1 and S^2 respectively, we now derive the relationship that unifies them.

Derivation of the Energetic Exchange Rate

Lemma 2.13 (DOF-Indifference). *By Isotropy Theorem (no privileged relational directions of state transformation 1.17) and the Ontological Minimalism Principle (minimum primitives, 1.3), each independent degree of freedom (DOF) has equal relational weight. WILL has no preferred directions and no preferred degrees of freedom.*

Proof. Any unequal weights to DOF's would create a preferred direction on maximally symmetric carriers violating the **Relational Isotropy Theorem** (1.17). It would require an extra rule that distinguishes the carriers on grounds other than their internal dimensions. Such a rule would constitute hidden structure, violating all four methodological principles (1.5, 1.3, 1.4, 1.2). Therefore, each independent DOF must receive an identical relational weight l . $\square$

Theorem 2.14 (Closure). *In a relationally closed system, the kinematic carrier S^1 (1 DOF) and the potential carrier S^2 (2 DOF) exchange the active relational ledger in proportion to their degrees of freedom. Consequently, any relationally closed system must globally satisfy*

$$\boxed{\text{Pull}^2 = 2 \times \text{Push}^2} \rightarrow \boxed{\kappa^2 = 2\beta^2}$$

Proof. By the Conservation Theorem (2.6) external amplitudes (β, κ) complete their ledger only in their quadratic forms ($\underbrace{\text{Amplitude}^2}_{\text{External Interaction}} + \underbrace{\text{Phase}^2}_{\text{Internal Existence}} = 1$). Let l be the active relational ledger allocated to one independent quadratic degree of freedom. By Lemma 2.13, every active quadratic DOF receives exactly this amount l .

- **Square amplitude on S^1 (1 DOF):** $\beta^2 = 1 \cdot l$
- **Square amplitude on S^2 (2 DOF):** $\kappa^2 = 2 \cdot l$
- **Substituting l immediately yields closure:** $\kappa^2 = 2\beta^2$

Any non-quadratic exchange law, same as any exchange rate other than 2 would require introduction of unjustified ontology either unequal weighting of degrees of freedom violating DOF-Indifference Lemma (2.13) or an additional free parameters violating all four principles (1.5, 1.3, 1.4, 1.2). Therefore:

$$\underbrace{1\text{Amplitude}^2}_{\text{Pull on } S^2} = \underbrace{2\text{Amplitude}^2}_{\text{Push on } S^1}$$

$\square$

Remark 2.15 (Carrier capacities fix the rate). *The exchange rate 2 exactly mirrors the ratio of the carriers' maximum absolute capacities. The one-DOF carrier S^1 saturates at $\beta_{\max}^2 = 1$ (light, $\beta_Y = 0$). The two-DOF carrier S^2 saturates at $\kappa_{\max}^2 = 2$ (the combined extremal-Kerr state $r = \frac{1}{2}R_s$, $\beta = 1$ as proved in (R.O.M. sec:Kerr)). Equal quadratic weight per DOF (Lemma 2.13) guarantees that the active ledger ratio ($\kappa^2/\beta^2 = 2$) is identical to the maximal ratio ($\kappa_{\max}^2/\beta_{\max}^2 = 2$).*

Definition 2.16 (Closure Factor).

$$\delta_o \equiv \frac{2\beta_o^2}{\kappa_o^2}$$

Here and throughout all RG papers κ, β are global and κ_o, β_o are orbital-phase-dependent local amplitudes (applied for all parameters): $\kappa^2 = \langle \kappa_o^2 \rangle$ and $\beta^2 = \langle \beta_o^2 \rangle$ For circular orbits, $\delta_o \equiv 1$. For elliptic orbits locally, $\delta_o \neq 1$ as demonstrated in R.O.M. eccentricity section .

As we demonstrate in R.O.M. addition paper this closure factor is the relational origin of orbital eccentricity:

Relational eccentricity	
Perihelion Push ($2\beta_p^2 > \kappa_p^2$)	$\frac{2\beta_p^2}{\kappa_p^2} - 1 \equiv e \equiv 1 - \frac{2\beta_a^2}{\kappa_a^2}$ Aphelion Pull ($2\beta_p^2 < \kappa_p^2$)
ECCENTRICITY $\equiv$ CLOSURE DEFECT	
SPACETIME $\equiv$ ENERGY	

Orbital geometry is the ratio of energy projections. This result supports the Unifying Ontological Primitive (

Corollary 2.17 (Relational Closure Condition). *Closed systems (momentary or periodic) satisfy $\kappa^2 = 2\beta^2$. Open systems display $\langle \delta \rangle_{\text{cycle}} \neq 1$, the magnitude of which quantifies the energy flow through unaccounted channels. When all channels are included, closure is restored.*

Illustrative Examples.

- **Circular Orbit (Closed).** A body at any orbital phase satisfies $\kappa_o^2 = 2\beta_o^2$. The entire active ledger is partitioned identically between kinetic Push and gravitational Pull projections; no internal geometric oscillation and no external channel exists.
- **Elliptical Orbit (Closed).** A body satisfies $\langle \kappa_o^2 \rangle = 2\langle \beta_o^2 \rangle$ as an average per orbital cycle due to the internal geometric oscillation ("breathing") of eccentric systems. This internal oscillation is restricted by the Energy-Symmetry Law (??) so that the difference $\beta^2 = \kappa_o^2 - \beta_o^2 = \text{constant}$ at any orbital phase (R.O.M. sec:vis-viva). No external channel exists.
- **Radiating Binary (Open).** An elliptical compact binary violates Closure $\langle \kappa_o^2 \rangle \neq 2\langle \beta_o^2 \rangle$ when only orbital degrees of freedom are counted. Including all channels restores Closure.

Remark 2.18 (Geometric Origin of Physical Law). *The relation between kinetic Push and potential Pull energy projections is a geometric necessity of ledger Closure. Aspects of WILL maintain coherence through the invariant DOF ratio. This is the relational origin of the **virial theorem**:*

$$\text{Potential Pull}^2(\kappa^2) \equiv 2 \times \text{Kinetic Push}^2(\beta^2).$$

Structural Independence of the Classical Force Law and Closure Coefficient

In classical mechanics, the virial coefficient is dynamically coupled to the spatial force law. For a power-law potential $V \propto r^n$, the general virial theorem dictates $2\langle T \rangle = n\langle V \rangle$. For Newtonian gravity ($n = -1$), this yields $2\langle T \rangle = -\langle V \rangle$, establishing the energy magnitude ratio of 2. In this legacy view, the force law dictates the virial coefficient; they are two sides of the same dynamical coin.

WILL Relational Geometry decouples them. The $1/r$ proportionality and the Closure coefficient 2 are proven to emerge from **independent geometric channels**:

- **Channel 1 (Potential–Distance):** The $\kappa^2 \propto 1/r$ relation is derived by demoting spatial distance from fundamental primitive to inverse Pull amplitude on S^2 carrier (2.11). The S^1 carrier and DOF ratios are entirely absent from this derivation.
- **Channel 2 (Potential–Kinetic Exchange):** The $\kappa^2 = 2\beta^2$ relation is derived from the topological DOF ratio (1 for S^1 , 2 for S^2) enforced by DOF-Indifference (2.13). Spatial distance r is entirely absent from this derivation.

Because both channels derive from the same underlying Relational Geometry, their mutual consistency is *geometrical*, not dynamical. WILL precludes the possibility of a $1/r$ gravitational potential exhibiting Closure coefficient other than 2. This structural identity replaces the need for postulated equations of motion, Lagrangians, or action principles as foundational primitives, providing additional confirmation to the results from section on classical mechanics (??).

Derivation Chain Map

$\downarrow$ 3 R as Philosophical Pillars: Rationalism, <i>Operational</i> Realism, Relationalism $\downarrow$			
Method: $\downarrow$ (<i>Epistemic Hygiene</i> + <i>Relational Origin</i> + <i>Ontological Minimalism</i> + <i>Math Transparency</i>) $\downarrow$			
Standard Primitives:		$\underbrace{\text{background manifold} + \text{metric}}_{\text{spacetime structure (not derived)}}$	$+ \underbrace{\text{G, c, M, fields}}_{\text{energy dynamics (postulated)}}$
$\downarrow$ (<i>Ontological reduction by Relational Origin and Minimalism Principles no background allowed</i>) $\downarrow$			
One Primitive: <i>WILL</i> $\equiv$ <i>SPACE-TIME-ENERGY</i> $\implies$		SPACETIME $\equiv$ ENERGY	
$\downarrow$ All physically meaningful quantities are relational features of WILL $\downarrow$			
<i>Topology of WILL:</i> Relational Closure + Causal Continuity + Isotropy Theorems			
$\downarrow$ <i>Geometry of WILL:</i> S^1 (1-DOF kinematic Push) and S^2 (2-DOF potential Pull) $\downarrow$			
Relational Conservation Theorem:		$\underbrace{\text{Amplitude}^2}_{\text{External Transformation}}$	$+ \underbrace{\text{Phase}^2}_{\text{Internal Change}} = 1$
$\underbrace{1/\beta_Y = 1 + z_\beta}_{\text{transverse Doppler shift}}$	$\underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{Kinematic Push } S^1}$	Relational Projections	$\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{Potential Pull } S^2 \text{ gravitational redshift}}$ $\underbrace{1/\kappa_X = 1 + z_\kappa}_{\text{gravitational redshift}}$
$\downarrow$ (<i>by DOF-Indifference Lemma: equal quadratic weight to each independent DOF</i>) $\downarrow$			
Closure Theorem:		$\underbrace{1 \text{ PULL}^2}_{\text{Amplitude on } S^2}$	$= \underbrace{2 \text{ PUSH}^2}_{\text{Amplitude on } S^1} \implies \boxed{\kappa^2 = 2\beta^2}$